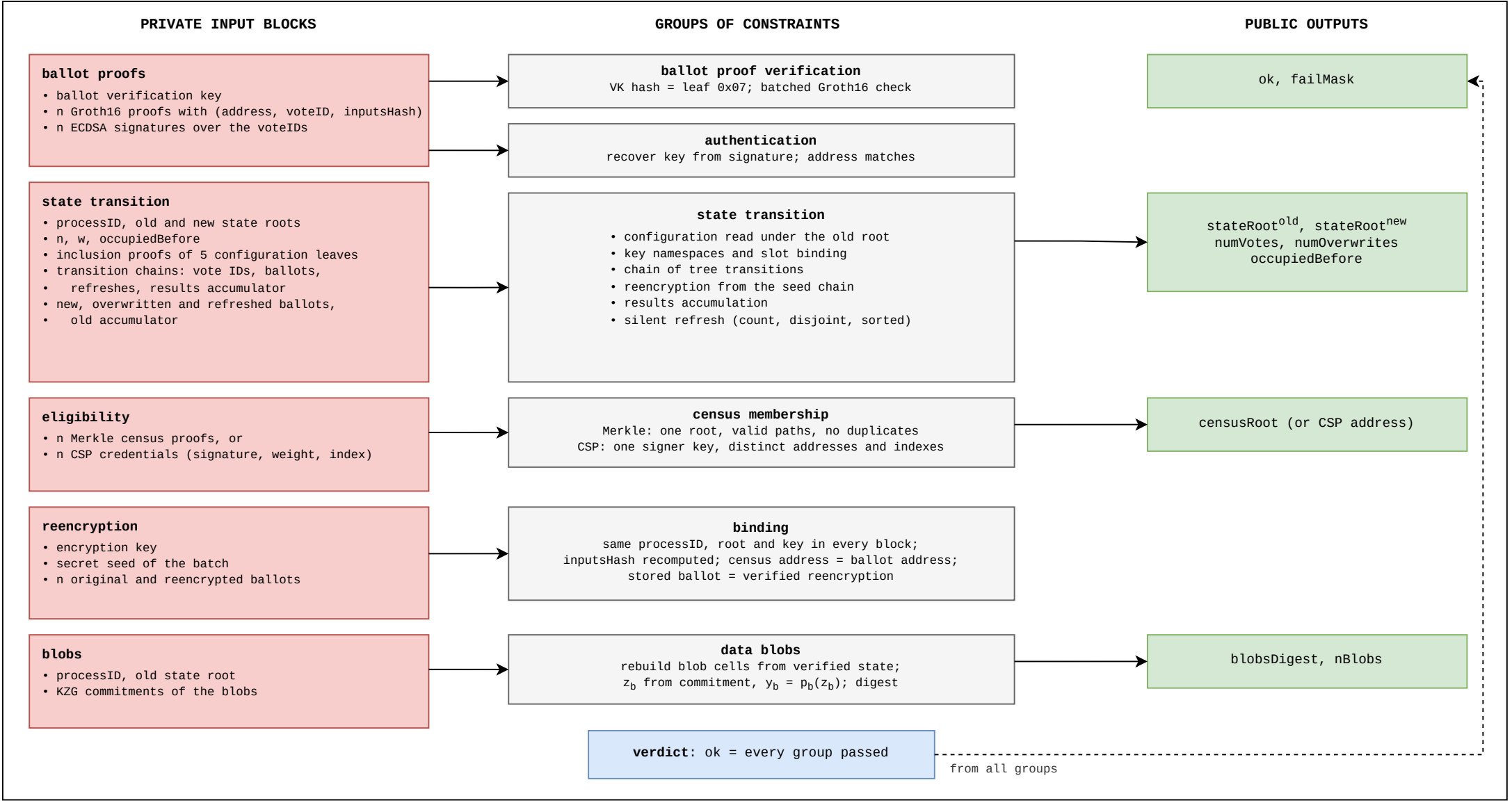
PRIVATE INPUT BLOCKS
GROUPS OF CONSTRAINTS
PUBLIC OUTPUTS
ballot proofs
• ballot verification key
• n Groth16 proofs with (address, voteID, inputsHash)
• n ECDSA signatures over the voteIDs
ballot proof verification
VK hash = leaf 0x07; batched Groth16 check
authentication
recover key from signature; address matches
ok, failMask
state transition
• processID, old and new state roots
• n, w, occupiedBefore
• inclusion proofs of 5 configuration leaves
• transition chains: vote IDs, ballots,
• refreshes, results accumulator
• new, overwritten and refreshed ballots,
• old accumulator
state transition
• configuration read under the old root
• key namespaces and slot binding
• chain of tree transitions
• reencryption from the seed chain
• results accumulation
• silent refresh (count, disjoint, sorted)
stateRoot^old, stateRoot^new
numVotes, numOverwrites
occupiedBefore
eligibility
• n Merkle census proofs, or
• n CSP credentials (signature, weight, index)
census membership
Merkle: one root, valid paths, no duplicates
CSP: one signer key, distinct addresses and indexes
censusRoot (or CSP address)
reencryption
• encryption key
• secret seed of the batch
• n original and reencrypted ballots
binding
same processID, root and key in every block;
inputsHash recomputed; census address = ballot address;
stored ballot = verified reencryption
blobs
• processID, old state root
• KZG commitments of the blobs
data blobs
rebuild blob cells from verified state;
z_b from commitment, y_b = p_b(z_b); digest
blobsDigest, nBlobs
verdict: ok = every group passed
from all groups